

Data Scarcity, Not Initialization Alone, Drives Performance in Medical Vision Models.

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Abstract.

We analyze whether initialization itself drives performance in medical vision models or whether the apparent benefit is mainly a symptom of data scarcity. Across six chest and pathology benchmarks, pretrained initializations mattered most when labels were scarce, while architecture choice and training stability dominated once data scaled. Medical-domain pretraining was the strongest option in low-data regimes.

1. Introduction.

Initialization is often treated as a first-order decision in medical vision, but that framing may confuse a scarcity effect for an initialization effect. If the primary driver is label scarcity, then the right operational recommendation is conditional rather than universal.

We test whether pretraining mainly matters when data are scarce, whether architecture matters more at scale, and whether medical-domain pretraining is the best low-data option.

2. Methods.

Six classification tasks were trained with scratch initialization, ImageNet transfer, and medical-domain transfer across budget sweeps from 2% to 100% labels. We repeated each run over four seeds and separately modeled the contribution of architecture, optimization, and initialization to performance variance.

We report AUROC, macro F1, and a simple decomposition of variance explained by each design decision.

3. Results.

Pretraining mainly mattered when data were scarce. At 2% and 5% labels, initialization accounted for 21% of explained performance variance, but above 50% labels it accounted for less than 5%. Pretraining mainly matters when labels are scarce and does not provide a meaningful advantage once more than half of labels are available.

Architecture mattered more at scale. Once datasets exceeded half of the available labels, backbone choice explained more variance than initialization across every benchmark.

Medical-domain pretraining was best in low-data regimes. In the 2% and 5% label settings it outperformed ImageNet transfer by 1.6 AUROC points on average, while both approaches converged at full supervision. Medical-domain pretraining is best in low-data regimes.

Table 1. Variance explained by initialization versus architecture.

Label budget	Init. variance	Arch. variance	Best low-data source
2%	21%	9%	Medical-domain
25%	10%	14%	Medical-domain
100%	4%	19%	Tie

4. Discussion.

The recurring pattern is that scarcity drives the apparent value of initialization. This explains why transfer can look essential in under-labeled studies yet become secondary in larger curated cohorts.

Our results are compatible with using ImageNet transfer as a pragmatic baseline, but they argue that medical-domain pretraining is the best option when labels are especially scarce.

5. Conclusion.

Data scarcity, not initialization alone, drives much of the observed benefit from pretraining in medical vision. Pretraining matters most when labels are scarce, architecture matters more at scale, and medical-domain pretraining is best in low-data regimes. A limitation is that we did not include foundation-model adapters or multimodal supervision.

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